POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

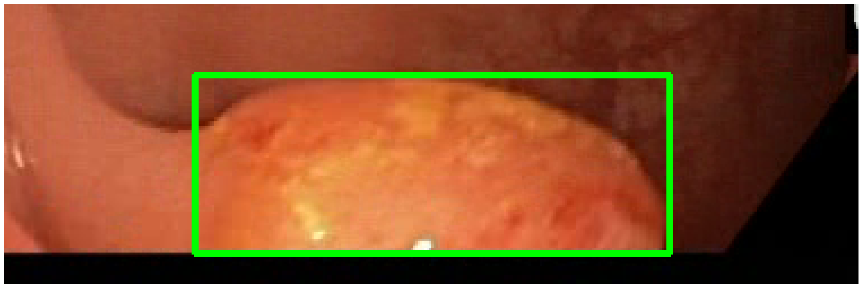
Video ID:	afbc5638-e8eb-454e-aab3-80f6f3922966
Total Frames:	8445
Frame Rate:	25.0 fps (assumed)
Duration:	337.8 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	2517	✓
RT-DETR Detections	3342	✓
MedSAM2 Detections	3600	✓
Consensus (3-Model Agreement)	4	✓
Risk Classification	**Count**	**Percentage**
HIGH RISK	4	100.0%
MEDIUM RISK	0	0.0%
LOW RISK	0	0.0%
TOTAL ANALYZED	4	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — MALIGNANT_POLYP [HIGH_RISK] (Tracked across 144 frames)

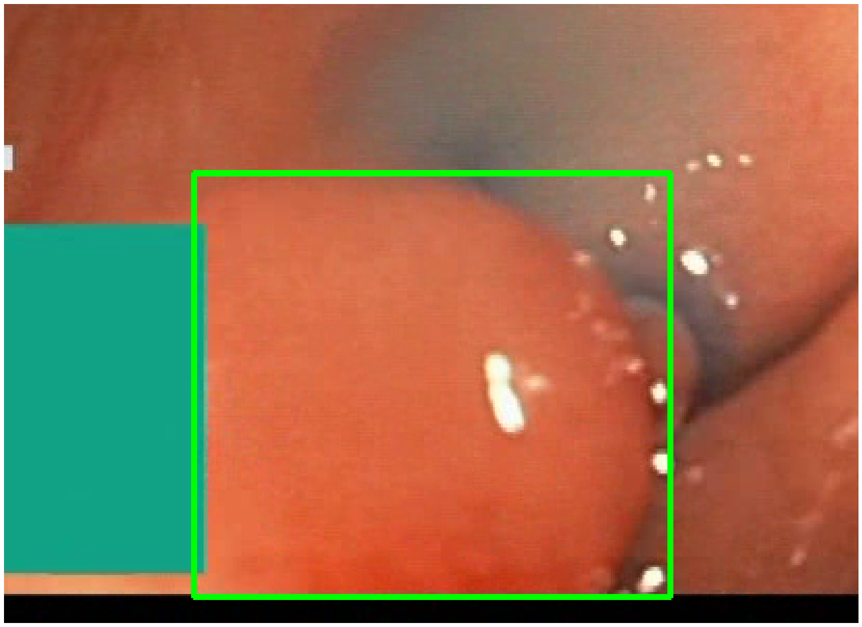


Feature	Value	Interpretation
Frame Index	1350	Frame 1350 of 8445
Detection Method	Detector Consensus (YOLO + RT-DETR agreement) (MedSAM2 unavailable for this video)	MedSAM2 Unavailable
POLYP TYPE	**MALIGNANT_POLYP**	Type confidence: 92.0%
Redness Score	0.329	Color-based vascularization
Relative Radius	22.1%	Polyp size as % of frame diagonal
Texture Score	0.245	Surface morphology
Vessel Visibility	0.438	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	LOW_RISK	Final symbolic reasoning bucket
Expert Prediction	LOW_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	87.5%	YOLO / RT-DETR / track confidence

Type Assessment: Morphology consistent with high-grade neoplasia — biopsy and urgent resection strongly recommended.

Low-confidence HIGH RISK detection - Further evaluation recommended

POLYP #2 — BLEEDING_POLYP [HIGH_RISK] (Tracked across 989 frames)

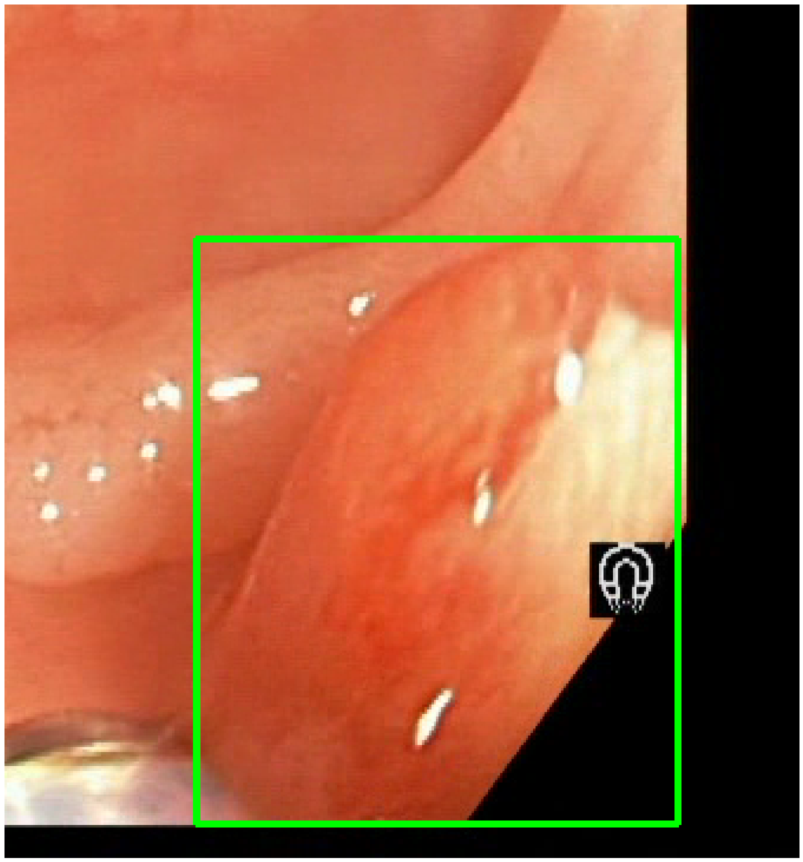


Feature	Value	Interpretation
Frame Index	3428	Frame 3428 of 8445
Detection Method	Detector Consensus (YOLO + RT-DETR agreement) (MedSAM2 unavailable for this video)	MedSAM2 unavailable
POLYP TYPE	**BLEEDING_POLYP**	Type confidence: 92.0%
Redness Score	0.238	Color-based vascularization
Relative Radius	19.6%	Polyp size as % of frame diagonal
Texture Score	0.061	Surface morphology
Vessel Visibility	0.885	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	87.1%	YOLO / RT-DETR / track confidence

Type Assessment: Active hemorrhagic lesion — elevated vascularity with redness signal. Immediate clinical review required.

Low-confidence HIGH RISK detection - Further evaluation recommended

POLYP #3 — MALIGNANT_POLYP [HIGH_RISK] (Tracked across 666 frames)



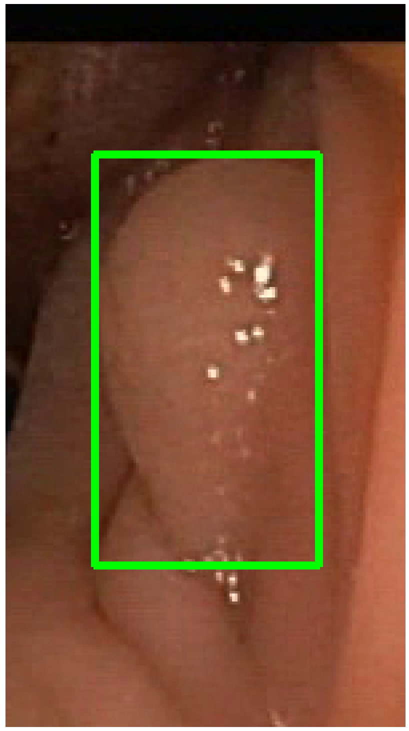
Feature	Value	Interpretation
Frame Index	3329	Frame 3329 of 8445
Detection Method	Detector Consensus (MedSAM2 Unavailable)	YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)
POLYP TYPE	**MALIGNANT_POLYP**	Type confidence: 92.0%
Redness Score	0.216	Color-based vascularization
Relative Radius	12.0%	Polyp size as % of frame diagonal
Texture Score	0.058	Surface morphology
Vessel Visibility	0.567	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification

Clinical Classification	LOW_RISK	Final symbolic reasoning bucket
Expert Prediction	LOW_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	88.3%	YOLO / RT-DETR / track confidence

Type Assessment: Morphology consistent with high-grade neoplasia — biopsy and urgent resection strongly recommended.

Low-confidence HIGH RISK detection - Further evaluation recommended

POLYP #4 — BLEEDING_POLYP [HIGH_RISK] (Tracked across 25 frames)



Feature	Value	Interpretation
Frame Index	7267	Frame 7267 of 8445
Detection Method	Detector Consensus (YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)	(MedSAM2 Unavailable)
POLYP TYPE	**BLEEDING_POLYP**	Type confidence: 92.0%
Redness Score	0.234	Color-based vascularization
Relative Radius	21.9%	Polyp size as % of frame diagonal
Texture Score	0.092	Surface morphology
Vessel Visibility	0.825	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	82.1%	YOLO / RT-DETR / track confidence

Type Assessment: Active hemorrhagic lesion — elevated vascularity with redness signal. Immediate clinical review required.

Low-confidence HIGH RISK detection - Further evaluation recommended

CLINICAL IMPRESSION

Summary:
A total of 4 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:
• HIGH RISK polyps: 4 (100.0%)
• MEDIUM RISK polyps: 0 (0.0%)
• LOW RISK polyps: 0 (0.0%)

Recommendation:
The presence of HIGH RISK polyps warrants close clinical attention. Consider endoscopic resection or follow-up surveillance depending on clinical context.

Technical Details:
• Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
• Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
• Classification: Cluster-specific Decision Tree experts
• Confidence: Based on deep learning + medical heuristics

Disclaimer: *This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.*

AI-Based Polyp Detection System
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